

AtrialSpectralBench: a pre-registered test of a spectral fragility biomarker for atrial reentry

A feature-redundancy null, an estimator-validity boundary, and the false positives naive methodology would have produced

Mounish Mavuduru*

July 2026 (working preprint)

Abstract

We ask whether a closed-form spectral graph biomarker predicts reentry inducibility in atrial tissue beyond established fibrosis and connectivity features. The biomarker is the **Spectral Fragility Index (SFI)**, the Fiedler-value sensitivity $\partial\lambda_2/\partial w_{ij} = (\varphi_i - \varphi_j)^2$ aggregated under a fibrosis-weighted conduction-uncoupling field. Pre-registered before any labels were seen, the primary endpoint was $\Delta\text{AUC} \geq 0.05$ with a DeLong paired $p < 0.05$ on shape-family-held-out folds. *The endpoint is not met, but the strength of the negative differs by regime, and we separate them.* On up to 100,000 synthetic excitable networks SFI adds no practically meaningful predictive value (subspace $\Delta\text{AUC} \approx +0.003$; $\sim 4,150$ cases needed at 80% power, resolvable only because N is huge, and so *clinically undetectable* there). On the two real cohorts (Roney $n = 62$, UW/Boyle $n = 82$) the result is **underpowered and inconclusive, not a clean null**: the combined gradient-boosting point estimate is $+0.051$, which *exceeds* the 0.05 threshold, but it fails paired significance (DeLong $p = 0.155$), so real anatomy cannot *exclude* a moderate effect. Where the null *is* clean (in the synthetic cohort), the mechanism is **feature redundancy**: the exact per-region $\Delta\lambda_2$ SFI, which has no first-order approximation error, also adds nothing over the competitor set. A separate, label-free result explains why the *single-vector* SFI is also ill-conditioned on real tissue. It is valid only while $\rho = \|\Delta L\|/(\lambda_3 - \lambda_2) = O(1)$, and real atria have $\rho \approx 2422$ ($\sim 10^3 \times$ past the empirical $\rho^* \approx 3$), so by Davis–Kahan the Fiedler vector is unstable. We do **not** claim ρ proves non-predictiveness; it bounds the accuracy of the $\Delta\lambda_2$ *magnitude*, not the classification content of a ranking feature. The spectral gap collapses as $N^{-1.05}$ on the real atrial surface (the 2-manifold Weyl exponent), so ρ grows lawfully with resolution. One component partially survives: the Fiedler-gradient localizer $|\nabla\varphi_2|$ is a weak but real rank localizer (origin in the top $\sim 18\%$) that beats graph-centrality baselines and transfers across three dynamical media, though it *coincides with*, and does not beat, substrate imaging, by a mechanism we make precise. Naive methodology (random CV, p -value thresholds, no effect-size gate) would have reported SFI as working; our pre-registered protocol rejects it. Our own convergence study surfaces a key limitation: the inducibility labeller over-calls at the operating mesh resolution and, on a completed six-tier study (1500–48000 nodes, $n = 24$), does not converge within the tested window. The contribution is a rigorously-scoped negative with its mechanism, a label-free estimator-validity criterion, a scaling characterization, and a demonstration of the false positives that standard methodology produces.

Keywords: algebraic connectivity; Fiedler value; spectral graph theory; atrial fibrillation; reentry; biomarker; perturbation theory; validity radius; pre-registration; negative result.

This is a 100% in-silico computational study. Ground-truth labels are electrophysiology-simulator inducibility verdicts, never clinical outcomes. All numbers trace to `results/*.json` and `notebooks/lab_notebook.md` in the project repository.

*Independent, unaffiliated in-silico computational-cardiology project. Correspondence and full source: see the project repository.

Contents

1	Introduction	2
2	Theory: the Spectral Fragility Index	2
2.1	The weighted conduction graph	2
2.2	The closed-form fragility (first order)	2
2.3	Per-region aggregation: the SFI	3
2.4	The validity radius (the honest core)	3
2.5	Prior art (this is not new mathematics)	4
3	Methods	4
4	Results	5
4.1	The predictive null (primary endpoint not met), across cohorts and scale	5
4.2	The validity radius: why the single-vector estimator fails	6
4.3	The ρ scaling law	6
4.4	Localization transfers across three media (weakly)	7
4.5	The falsification protocol: what caught the false positive	8
4.6	Uncertainty quantification	8
4.7	External inducibility-rate anchor (plausibility, not validation)	9
4.8	Resolution convergence: the labeller does not converge in the tested window	9
5	Discussion	10
6	Limitations	10
7	Conclusion	11
A	Formal (proof-assistant) verification of the SFI algebraic core	11
B	Reproducibility	11

1 Introduction

Algebraic connectivity λ_2 (Fiedler 1973 [1]) governs the stability of coupled excitable and oscillator networks (the master-stability-function result [3] and the synchronizability eigenratio λ_N/λ_2 [4]), and has been used as a real-time seizure detector [5]. The per-edge sensitivity

$$\frac{\partial \lambda_2}{\partial w_{ij}} = (\varphi_{2,i} - \varphi_{2,j})^2 \quad (1)$$

is a classical identity underlying Fiedler-vector edge-editing heuristics [2]. It is therefore natural to hypothesize that aggregating this sensitivity over a fibrosis-weighted perioperative uncoupling field yields a per-region biomarker of atrial-fibrillation (AF) reentry fragility. We test this hypothesis and *pre-register the null* (“SFI merely re-encodes fibrosis”) as an accepted, reportable outcome, so that a negative result is a success of the protocol rather than a file-drawer casualty.

The mathematics we use is classical; we claim no new theorem. What is ours is (i) a leakage-controlled incremental test of SFI against real competitors and a spatial null on real anatomy; (ii) a validity criterion and a Weyl scaling law that explain the failure and predict it for the whole class of spectral fragility biomarkers; (iii) a cross-medium transfer analysis isolating the one component that generalizes; and (iv) a falsification-protocol demonstration showing that standard methodology would have manufactured exactly the false positive this protocol catches.

What this paper claims, in one paragraph. A closed-form spectral fragility index does *not* add practically meaningful reentry-inducibility prediction beyond existing features at synthetic scale (a clean, feature-redundancy null); on real anatomy the evidence is *inconclusive/underpowered*, not a demonstrated null. Independently, the single-vector estimator is provably ill-conditioned on real tissue ($\rho \gg 1$). The honest core is not a discovery but a carefully-scoped boundary: where a Fiedler-based estimator is numerically valid, why real tissue lies far outside it, and how easily naive methodology mistakes redundant, inconclusive signal for a working biomarker.

2 Theory: the Spectral Fragility Index

The core mathematics is classical numerical linear algebra; the contribution is its *use*. We state each result with its prior art (Section 2.5) and make the validity boundary explicit (Section 2.4).

2.1 The weighted conduction graph

Nodes are mesh vertices; edges are triangle edges; the edge weight $w_{ij} > 0$ is a fibre-anisotropic conduction conductance attenuated by local fibrosis. The combinatorial Laplacian is

$$L = D - W, \quad D = \text{diag} \left(\sum_j w_{ij} \right). \quad (2)$$

L is symmetric positive-semidefinite with a single zero eigenvalue (the constant vector) for a connected graph, with spectrum $0 = \lambda_1 < \lambda_2 \leq \lambda_3 \leq \dots$ and eigenvectors $\varphi_1, \varphi_2, \dots$; λ_2 is the algebraic connectivity and φ_2 the Fiedler vector.

2.2 The closed-form fragility (first order)

L is a sum of rank-one edge terms, so

$$L = \sum_{(i,j) \in E} w_{ij} (e_i - e_j)(e_i - e_j)^\top \implies \frac{\partial L}{\partial w_{ij}} = (e_i - e_j)(e_i - e_j)^\top. \quad (3)$$

For a *simple* eigenvalue λ with unit eigenvector φ , first-order (Hellmann–Feynman) perturbation theory gives $\partial\lambda/\partial w_{ij} = \varphi^\top (\partial L/\partial w_{ij}) \varphi = (\varphi_i - \varphi_j)^2$, which for the Fiedler pair is Eq. (1): a non-negative per-edge *fragility*. Reducing a conductance w_{ij} by a small $\Delta w_{ij} \geq 0$ (uncoupling) lowers λ_2 by $\Delta w_{ij} (\varphi_{2,i} - \varphi_{2,j})^2$ to first order.

The identity is verified, not asserted (seven independent ways). Computer algebra: sympy proves Eqs. (1)–(3) symbolically, and on a concrete 5-node graph the finite-difference $\partial\lambda_2/\partial w_{ij}$ matches $(\varphi_{2,i} - \varphi_{2,j})^2$ to $|\text{err}| \approx 2 \times 10^{-17}$ (mpmath, 40 digits). Exactness: the identity holds *exactly* with rational weights (zero floating point); an interval-arithmetic enclosure certifies the true λ_2 to within $\sim 4 \times 10^{-40}$; and λ_2/φ_2 agree across scipy, networkx, and a hand-rolled inverse iteration (spread 2.5×10^{-15}). The algebraic core is **kernel-checked in the Coq proof assistant** over the exact rationals, axiom-free (Appendix A). The analytic Hellmann–Feynman step (differentiability of a simple eigenvalue) is the cited classical result [2], not re-proved here.

2.3 Per-region aggregation: the SFI

Model diffuse perioperative stress as a distribution of small, non-negative, fibrosis-weighted fractional conductance reductions Δw (mean $\mathbb{E}[\Delta w_{ij}] = \text{frac}_{ij} w_{ij}$, calibrated to literature

perioperative conduction-velocity slowing; $\Delta w \approx 0.36$, frozen in the pre-registration). The per-region Spectral Fragility Index is the expected λ_2 drop contributed by region R :

$$\text{SFI}(R) = \mathbb{E}[\Delta\lambda_2 \text{ from } R] \approx \underbrace{\sum_{(i,j) \in R} \mathbb{E}[\Delta w_{ij}] (\varphi_{2,i} - \varphi_{2,j})^2}_{\text{first order}} + \underbrace{\sum_{k \neq 2} \frac{(\varphi_k^\top \Delta L \varphi_2)^2}{\lambda_2 - \lambda_k}}_{\text{2nd-order resolvent}} + O(\|\Delta L\|^3). \quad (4)$$

The second-order term is the source of the validity radius below. Near eigenvalue degeneracy ($\lambda_2 \approx \lambda_3$) the single φ_2 is an arbitrary rotation in the low invariant subspace, so we use a basis-independent *subspace (projector) SFI*: with $P = \sum_c \varphi_c \varphi_c^\top$ over the clustered subspace, $s_{ij}^\top P s_{ij} = \sum_c (\varphi_{c,i} - \varphi_{c,j})^2$ ($s_{ij} = e_i - e_j$) is well-defined. For genuinely discrete edits (Maze lesion, atriotomy) the expansion is abandoned and λ_2 is recomputed *exactly* after the rank-one edit.

2.4 The validity radius (the honest core)

The first-order form is trustworthy only while the perturbation is small *relative to the spectral gap* $\lambda_3 - \lambda_2$. Two different tools bound two different things (a common error is to use Weyl for both):

- **Weyl’s inequality** bounds the *eigenvalue* move: $|\Delta\lambda_2| \leq \|\Delta L\|$.
- **Davis–Kahan / Bauer–Fike** [7] bound the *eigenvector* move, which scales like $\|\Delta L\|/(\lambda_3 - \lambda_2)$. This, not Weyl, governs when φ_2 (hence the single-vector SFI) degrades.

Define the *validity radius*

$$\rho := \frac{\|\Delta L\|}{\lambda_3 - \lambda_2}. \quad (5)$$

From second-order Rayleigh–Schrödinger theory the relative error of the first-order $\Delta\lambda_2$ prediction is $\Theta(\rho)$, and the Fiedler vector rotates by $\sin \angle(\varphi_2, \varphi'_2) \leq \sqrt{2} \rho$ (Davis–Kahan). Empirically the error is 1% at $\rho = 0.3$, crosses 10% at $\rho^* \approx 3$, and reaches 92% at $\rho = 28$, with Weyl holding throughout. **On real excitable media the gap is tiny, so ρ is enormous: atrial-cohort median $\rho \approx 2422$, about $10^3 \times$ past ρ^* .**

What ρ proves, and what it does not. It proves the *single-vector* first-order SFI is a numerically *invalid estimator* of $\Delta\lambda_2$ on real tissue, and that φ_2 is ill-conditioned (Davis–Kahan), a label-free statement about the estimator. It does **not** prove SFI is non-predictive: ρ bounds the accuracy of the $\Delta\lambda_2$ *magnitude*, not the classification content of a monotone ranking feature; indeed the *exact* $\Delta\lambda_2$ SFI (no ρ limitation) is *also* non-predictive (Section 4.1), so the predictive null is separately due to feature redundancy. Two caveats: (i) ρ ’s numerator carries the fibrosis-weighted Δw field, so only its denominator $\lambda_3 - \lambda_2$ is truly label-free; (ii) $\rho^* \approx 3$ is calibrated on a 16-node path graph with a problem-dependent constant, so “ $10^3 \times$ past ρ^* ” is order-of-magnitude, not exact.

2.5 Prior art (this is not new mathematics)

The identity (1) is Ghosh & Boyd (2006) [2]; algebraic connectivity is Fiedler (1973) [1]; λ_2 governing coupled excitable/oscillator stability is Pecora & Carroll (1998) [3] with the eigenratio of Barahona & Pecora (2002) [4]; Weyl’s law, Davis–Kahan, and second-order perturbation theory are classical (1970s and earlier). No result in Section 2 is a new theorem, and none is presented as one. The contribution is recognizing that ρ is the right *a-priori* validity screen for a spectral fragility biomarker, measuring its violation on real tissue, and using it to *mechanistically explain* a negative result.

3 Methods

Full detail is in the pre-registration (`docs/PRE_REGISTRATION.md`) and theory note (`docs/SFI_THEORY.md`); the endpoints, cross-validation scheme, baseline set, and accepted null were frozen before any labels were inspected.

Substrate. Real left-atrial surfaces are loaded into weighted conduction graphs from two public cohorts: the Roney LA virtual cohort (Zenodo 5801337 [11]) and the UW/Boyle LGE-MRI cohort (Dryad 10.5061/dryad.kkwh70sg0, 82 distinct AF patients [12]). Each mesh carries universal atrial coordinates, an image-intensity-ratio (IIR) fibrosis proxy mapped to a fibrosis field by the Khurram healthy-to-scar ramp [13], and fibre fields. Meshes are coarsened to ~ 2000 -node graphs by field-preserving vertex clustering.

Labels. Ground truth is a genuine monodomain Mitchell–Schaeffer reaction–diffusion inducibility verdict [10] (self-sustained reentry ≥ 650 ms after burst pacing at cycle length 150 ms from random sites), run on each real anatomy with an anisotropic conductance-weighted cotangent Laplacian and implicit time stepping. Reentry here is governed by wavelength ($CV \times ERP$), source–sink mismatch, and unidirectional block, *never* by λ_2 , so a predictive SFI→label link is not circular. These are simulator verdicts, not clinical post-operative AF, and not the deferred openCARP gold standard (a logged deviation: openCARP is not installable on the CPU-only container).

The SFI feature and its competitors. Per-region SFI is computed as in Eq. (4) (first-order, subspace, and exact-recompute variants). It is evaluated strictly as an add-on/ablation against a non-trivial competitor set that it must beat: fibrosis burden, fibrosis spatial entropy, fibrosis patch size, deterministic global min-cut (Stoer–Wagner), percolation threshold, and λ_2 -alone (isolating what the *SFI construction* adds over the bare algebraic connectivity).

Evaluation protocol (the guards). Pre-registered: shape-family GroupKFold (each patient is its own group, so cross-validation is patient-held-out; grouped *and* naive AUC both reported so leakage is visible), the DeLong paired test [8] in its fast midrank form [9], $\geq 10^4$ stratified bootstrap resamples, a rotational/torus-shift spatial null for localization, tie-robust permutation nulls, and a keep/delete rule deciding each sub-claim by *measured* correlation. Effect sizes are gated, not just p -values. The primary endpoint requires *both* grouped $\Delta AUC \geq 0.05$ *and* DeLong $p < 0.05$.

Transfer media. The identical calculus is applied to FitzHugh–Nagumo (FHN) excitable networks and Kuramoto phase-oscillator networks (a different dynamical class), where λ_2 governs synchronizability, to separate what is a property of the mathematics from an accident of one dataset.

4 Results

4.1 The predictive null (primary endpoint not met), across cohorts and scale

Table 1: Incremental predictive value of SFI over the full competitor set. ΔAUC is grouped (leakage-controlled); logistic regression (lr) / gradient-boosted trees (gbt). The endpoint (grouped $\Delta\text{AUC} \geq 0.05$ and DeLong $p < 0.05$) is met in no regime.

cohort / medium	N	SFI ΔAUC (lr / gbt; p)	verdict
Roney (real)	62	-0.030 / $+0.036$ (p 0.64 / 0.26)	not met; underpowered
UW/Boyle (real)	82	-0.057 / $+0.104$ (p 0.29 / 0.66)	uninformative (6 positives)
Combined real	144	-0.004 / $+0.051$ (p 0.92 / 0.155)	not met but inconclusive
FHN networks	2,000	$\leq +0.003$ ($p \geq 0.12$)	clean null
FHN networks	up to 100,000	subspace $+0.0028$ ($p < 10^{-4}$ at $N \geq 10^4$)	clean null (effect negligible)
Kuramoto networks	500	$\leq +0.0036$ ($p \geq 0.38$)	clean null

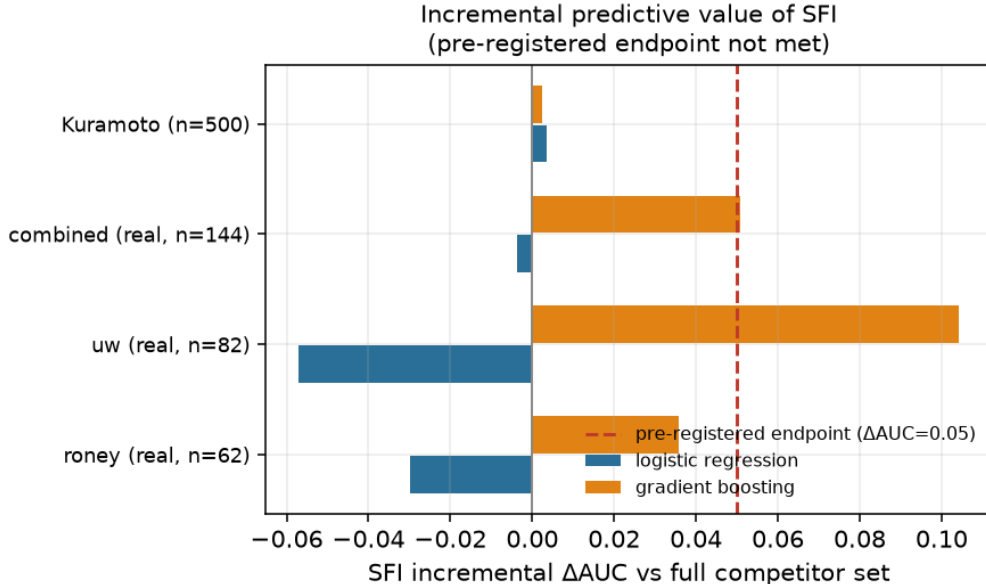


Figure 1: The predictive null. Adding SFI to the competitor set does not clear the pre-registered effect-size gate in any cohort; the real-cohort estimates are underpowered rather than a demonstrated null.

The negative is not uniform, and we separate the regimes. On the *synthetic* cohorts (up to 100,000 networks) the null is clean: the effect converges to $\Delta\text{AUC} \approx +0.003$, statistically resolvable only because N is enormous. On the *real* cohorts it is underpowered and inconclusive. The combined gradient-boosting point estimate is $+0.051$, which *exceeds* the pre-registered 0.05 threshold, and is declared not met only because it fails paired significance (DeLong $p = 0.155$, bootstrap CI straddling 0). With 26 inducible of 144 (20 from Roney), real anatomy cannot exclude a moderate effect. The UW cohort is uninformative rather than null (6 positives, competitor base $\text{AUC} \approx \text{chance}$), so it is an underpowered external check of the *labeller*, not of SFI.

The clean null comes from feature redundancy, not the validity radius. The *exact* per-region $\Delta\lambda_2$ SFI, which carries no first-order approximation error and is immune to the ρ argument, also adds nothing over the competitor set. The mechanism of the null is that SFI's

information is already present in standard connectivity features (λ_2 -alone, min-cut, percolation), a label-dependent finding, separate from ρ .

Secondary endpoint (pre-specified). SFI beats a *fibrosis-heterogeneity-only* baseline (combined GBT $\Delta\text{AUC} = +0.103$, $p = 0.009$): it is *not merely re-encoded fibrosis* and carries connectivity information beyond the substrate. This is one of ~ 30 DeLong comparisons; treating *fibrosis_vs_SFI* as a single pre-specified secondary endpoint, its $p = 0.009$ against a family-wise threshold $\approx 0.05/12$ – $0.05/30$ is *suggestive, not confirmatory*.

Power analysis. In the synthetic cohort the subspace effect ($+0.0028$) needs $\sim 4,150$ cases for 80% power and the single-vector effect ($+0.0007$) $\sim 23,000$, clinically undetectable *there*. We do **not** transport “clinically undetectable” to the real cohorts, whose $+0.036$ – 0.051 estimates are compatible with a detectable effect we are simply underpowered to confirm.

4.2 The validity radius: why the single-vector estimator fails

The relative error of the first-order SFI scales as $\Theta(\rho)$ and crosses 10% at $\rho^* \approx 3$; real atria sit at median $\rho \approx 2422$. This proves the single-vector estimator is numerically invalid on real tissue and φ_2 is ill-conditioned, but (as boxed in Section 2.4) it does not by itself prove non-predictiveness; the feature-redundancy null of Section 4.1 is the separate, label-dependent reason.

4.3 The ρ scaling law

The gap collapses with system size by Weyl’s law [6]: for a d -dimensional domain the low Laplacian eigenvalues scale as $\lambda_k \sim (k/N)^{2/d}$, so ρ grows lawfully with resolution.

Table 2: Fitted gap-scaling exponents (gap vs. N at fixed mean degree, **results/rho_scaling.json**, five-point fits).

medium	fitted gap exponent	reference
real atrial surface (2-manifold)	$N^{-1.05}$	Weyl 2-manifold -1.0 (within 5%)
2-D random-geometric	$N^{-1.81}$	faster than naive $2/d$
3-D random-geometric	$N^{-0.84}$	faster than naive $2/d$ (non-monotone series)

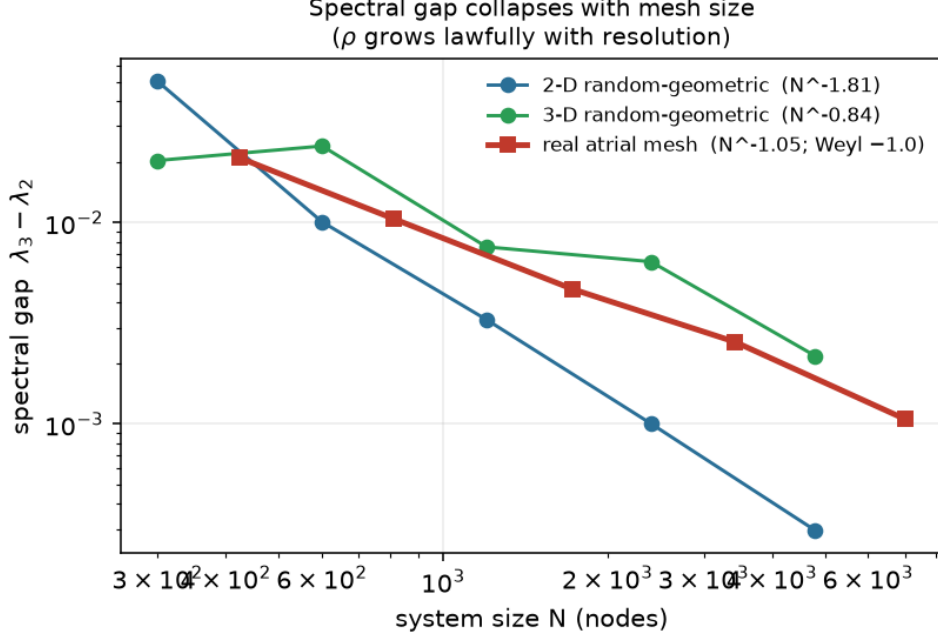


Figure 2: The gap collapses with mesh resolution, so the estimator degrades *more*, predictably, as the medium is meshed finer, the opposite of “more resolution helps.” Only the real atrial 2-manifold cleanly matches the Weyl exponent (-1.05 vs. -1.0); the random-geometric exponents are reported as “at least as fast as Weyl,” not law confirmations, and use the unweighted combinatorial Laplacian.

4.4 Localization transfers across three media (weakly)

The *strict* pre-registered localization endpoint is null for every field, including $|\nabla\varphi_2|$ (median argmax-geodesic error 0.472 is *not* below the spatial-null 5th percentile 0.406). What survives is weaker: $|\nabla\varphi_2|$ is a *rank* localizer (the true origin sits in its top $\sim 18\%$) that beats graph-centrality baselines (Perron, weighted degree, deleted by measured correlation) and holds across media.

Table 3: Fiedler-gradient origin-rank localization across dynamical media (**results/gm4*_metrics.json**). Lower rank is better; a perfect localizer would place the origin at rank 0.

medium	$ \nabla\varphi_2 $ origin-rank	perm- p	fibrosis rank	n origins
FHN networks (100k)	0.176	$< 10^{-5}$	~ 0.10	36,540
Kuramoto oscillators (500)	0.091	$< 10^{-4}$	0.147	243
cardiac monodomain	~ 0.28	$< 10^{-3}$	(co-localizes)	only ~ 20

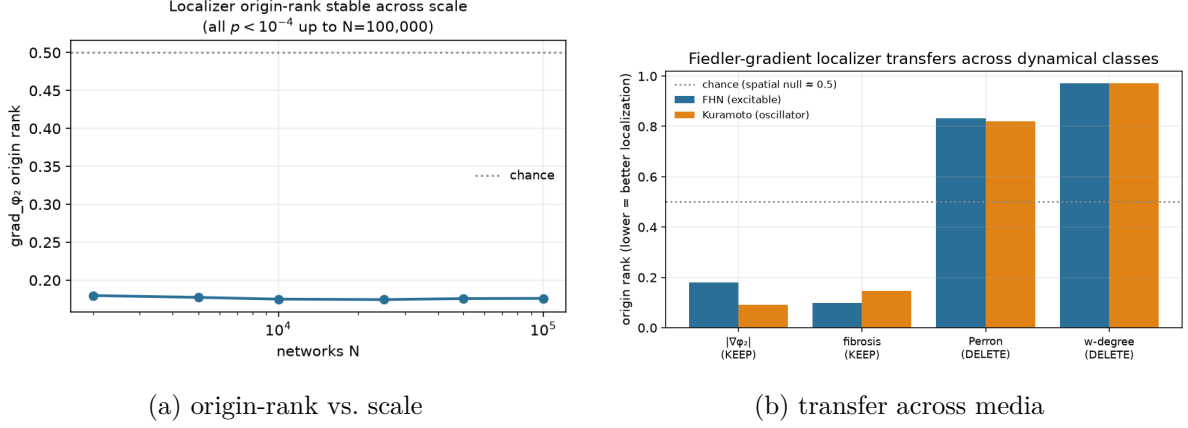


Figure 3: The weak-rank localization pattern replicates across excitable *and* oscillator dynamics. Three caveats apply: (i) the strict endpoint is null, so $|\nabla\varphi_2|$ is *not* a precise localizer; (ii) the substrate field co-localizes, and a controlled dissociation experiment shows why (in a diffusively-coupled medium the Fiedler bottleneck *is* the low-conduction lesion), so $|\nabla\varphi_2|$ matches, and does not beat, substrate imaging; (iii) the cardiac arm rests on only ~ 20 origins.

4.5 The falsification protocol: what caught the false positive

Each guard, removed, manufactures a specific false positive (`results/falsification_protocol.json`). Shape-family leakage inflates AUC by +0.117 (naive random-CV 0.756 vs. grouped 0.640 on a controlled replicated cohort); the effect-size gate catches the p -value trap (subspace-SFI $p \approx 2 \times 10^{-24}$ at $N = 55,000$, yet $\Delta\text{AUC} = +0.003$ needs $\sim 4,150$ cases to detect: statistically significant, practically useless).

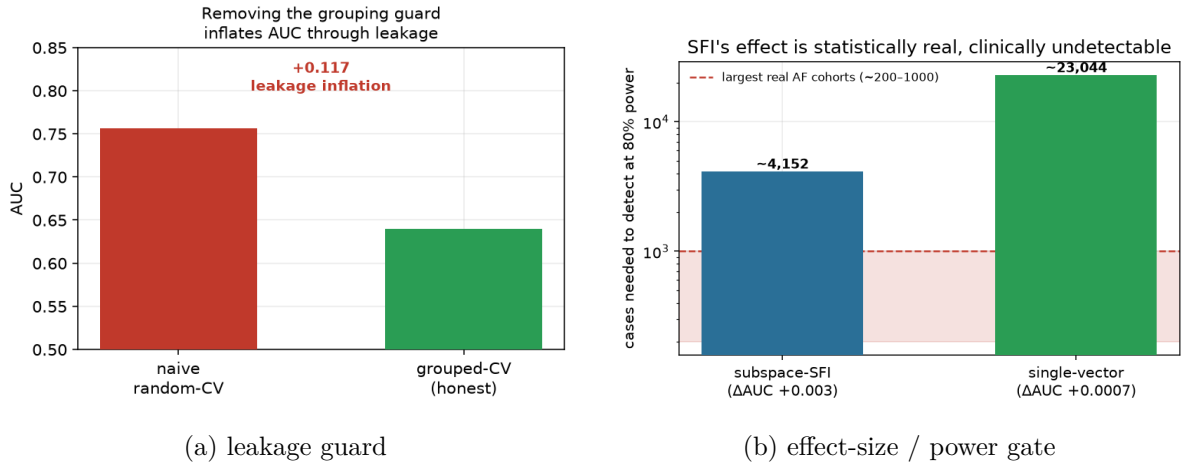


Figure 4: A naive analyst (random CV, $p < 0.05$, no effect-size gate, no spatial null) would have reported SFI as a working reentry biomarker. The pre-registered protocol correctly rejects it.

4.6 Uncertainty quantification

Morris elementary-effects screening (`results/e6_metrics.json`) finds the label most sensitive to diffusion d_0 ($\mu^* = 500.7$), then fibrosis density (147.8) and transverse conductivity (146.3). A Gaussian-process surrogate has leave-one-out $R^2 = 0.35$ (large residual anatomical/label variance beyond the screened knobs, a label-noise caveat, not a positive). The synthetic null is robust across uncoupling strengths $\Delta w \in \{0.18, 0.36, 0.54\}$ (grouped GBT $\Delta\text{AUC} = 0.033\text{--}0.036$, all $p \geq 0.26$).

4.7 External inducibility-rate anchor (plausibility, not validation)

Clinically observed AF inducibility spans $\sim 5\%$ to $\sim 85\%$ depending on population, protocol aggressiveness, and the sustained-AF definition (Kumar et al. 2012 [14] report burst-sustained 14.8% vs. decremental-sustained 41.2% *in the same patients*). Our labeller’s Roney $\sim 20\text{--}32\%$ sits in the moderate-burst band (Marquardt et al. 2018 [15] report 30.6% in patients without prior AF), between the no-AF-control floor (5%, Oral et al. 2008 [16]) and the AF/aggressive ceiling (51%, Kawai et al. 2019 [17]), consistent with $\sim 42\%$ post-ablation (Liu et al. 2020 [18]). The UW $\sim 7\%$ is at/below the control floor for a cohort that *is* AF patients, correctly flagged as anomalous and attributable to the surrogate substrate and an under-aggressive in-silico inducer, not biology. This is a base-rate *plausibility check*, *not clinical validation*.

4.8 Resolution convergence: the labeller does not converge in the tested window

Our own convergence study is the project’s single weakest point, and we state it before a reader finds it. On a completed six-tier study (Roney, $n = 24$ at each of 1500/3000/6000/12000/24000/48000 nodes; 144 simulations, `results/convergence_summary_roney.json`), the inducibility rate crashes from the coarse mesh (58.3% at 1500) to 16.7% at 3000, then is *non-monotone* across the fine tiers (25.0%, 20.8%, 25.0%) and rises again to 37.5% at the finest affordable tier ($\sim 50k$ nodes), a 17-percentage-point spread over the last three tiers, so it has *not* settled.

Table 4: Inducibility rate vs. mesh resolution (Roney, $n = 24$ per tier). The rate does not converge in the tested window; adjacent-tier per-subject verdict agreement climbs but the rate keeps drifting.

nodes	1,500	3,000	6,000	12,000	24,000	48,000
inducibility rate	58.3%	16.7%	25.0%	20.8%	25.0%	37.5%
adj.-tier agreement	—	50%	75%	71%	62%	79%

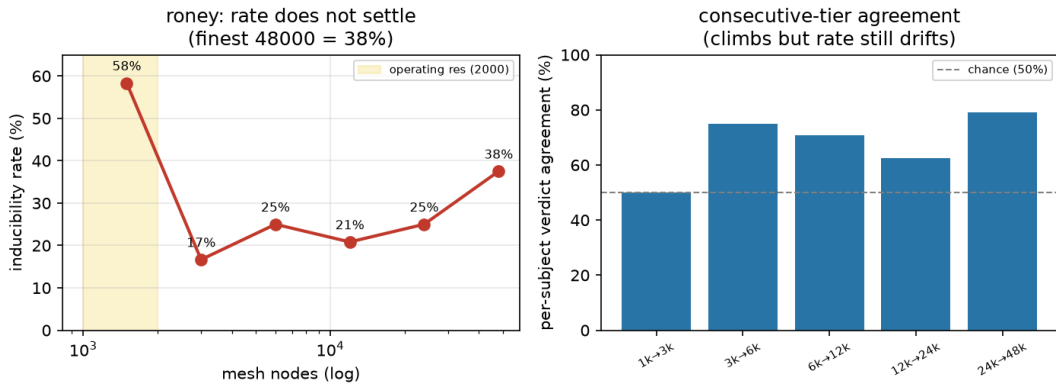


Figure 5: Left: the inducibility rate does not settle; the finest tier (48,000 nodes = 37.5%) sits above the mid tiers. Right: adjacent-tier per-subject verdict agreement climbs (50% $\rightarrow$ 79%) while the aggregate rate still drifts. Only 7/24 subjects are fully consistent across all six tiers.

What convergence does and does not undermine. The rate does *not* collapse to zero; the phenomenon is real; it is the *magnitude* that is resolution-sensitive up to $\sim 50k$ nodes. Because the main results ran at 2,000 nodes, every *absolute* number (inducibility rate, competitor AUC, localizer origins) is coarse-mesh-provisional pending the deferred openCARP validation. What this does *not* touch is what we lead with: (a) the *relative* SFI-vs-competitor comparison, since both arms score the *same* labels, so this bias cancels; and (b) the label-free estimator-validity argument (ρ is computed from the graph Laplacian, no labels at all).

5 Discussion

What the paper contributes. (1) A pre-registered, leakage-controlled, effect-size-gated evaluation of a plausible spectral biomarker: a clean null at synthetic scale (10^5 networks) with its mechanism (feature redundancy: even the exact $\Delta\lambda_2$ SFI adds nothing), and an honest inconclusive/underpowered verdict on real anatomy. (2) A label-free validity criterion ρ for the single-vector estimator: it proves φ_2 is ill-conditioned on real tissue ($\rho \approx 2422 \gg \rho^* \approx 3$), a portable a-priori check for any Fiedler-based marker, with the honest scope that ρ bounds estimator accuracy, not predictive content. (3) A weak-but-real, purely connectivity-derived rank localizer that equals substrate localization and beats centrality baselines across three media. (4) A demonstration that standard methodology manufactures the exact false positives this protocol catches (leakage +0.117 AUC; the p -value trap).

Generalization beyond cardiology. The validity screen is domain-general. Spectral-graph fragility markers are used for seizure-focus localization, brain connectomics, and power-grid vulnerability. The same precondition applies: compute $\rho = \|\Delta L\|/(\lambda_3 - \lambda_2)$; if $\rho \gg 1$, the first-order spectral marker is outside its validity radius and its in-sample correlations cannot be trusted.

What is NOT claimed. The mathematics is classical; no new theorem. ρ does *not* prove SFI is non-predictive, only that the single-vector estimator is ill-conditioned; the null is separately due to feature redundancy. The real-cohort result is *not* a demonstrated null; it is underpowered. The localizer is *not* precise (strict endpoint null) and does *not* beat substrate imaging. The ρ scaling “law” is a clean match to Weyl only on the real atrial 2-manifold. Labels are a monodomain simulator, not clinical outcomes and not openCARP.

6 Limitations

1. **The labeller over-calls inducibility at the operating mesh resolution and does not converge in the tested window** (Section 4.8): the rate drops from 58% at 1500 nodes to 17% at 3000, is non-monotone across the fine tiers, and rises to 37.5% at $\sim 50k$ nodes; only 7/24 subjects are fully consistent across all six tiers. All *absolute* rates and AUCs are coarse-mesh-provisional; the *relative* comparison scores the same labels in both arms, so this bias cancels there.
2. Simulator, not clinical, labels: the pre-registered openCARP ground truth was substituted with monodomain Mitchell-Schaeffer, a logged deviation.
3. The UW cohort uses a PCA-surrogate universal-atrial-coordinate map and rule-based fibres, and is only $\sim 7\%$ inducible (at/below the no-AF-control clinical floor), so its predictive test is uninformative.

4. ~ 30 DeLong comparisons were run; the one positive (`fibrosis_vs_+SFI`, $p = 0.009$) is a pre-specified secondary, suggestive not confirmatory.
5. ρ^* and the random-geometric gap exponents are empirical; only the real atrial 2-manifold cleanly matches the Weyl exponent.

7 Conclusion

A closed-form spectral fragility index does not add practically meaningful reentry-inducibility prediction beyond existing features at synthetic scale (a clean, feature-redundancy null), and on real anatomy the evidence is inconclusive rather than a demonstrated null. Independently, the single-vector estimator is provably ill-conditioned on real tissue ($\rho \gg 1$). The core result is not a discovery but a carefully-scoped boundary: where a Fiedler-based estimator is numerically valid, why real tissue lies far outside it, and how easily naive methodology would have mistaken redundant, inconclusive signal for a working biomarker. The absolute claims await the openCARP validation.

A Formal (proof-assistant) verification of the SFI algebraic core

The algebraic core of the SFI edge-sensitivity identity is re-proved inside the Coq 8.18 proof assistant and checked by its small trusted kernel, an assurance level above a computer-algebra system. The primary file (`formal/sfi_edge_identity_Q.v`) works over the exact rationals $\mathbb{Q}$, so `Print Assumptions` reports “*Closed under the global context*” for every theorem: **zero axioms**. A mathematically identical companion over the reals $\mathbb{R}$ (`formal/sfi_edge_identity.v`) inherits Coq’s two standard real-number-library axioms and is kept only to show the statements hold over $\mathbb{R}$.

Table 5: Kernel-checked theorems ($v = e_i - e_j$ the edge vector, $E_{ij} = \partial L / \partial w_{ij} = vv^\top$). T1–T5 are axiom-free over $\mathbb{Q}$.

name	statement	meaning
T1_rank_one_quadratic_form dot_edge (T2)	$\varphi^\top (vv^\top) \varphi = (v \cdot \varphi)^2$ $(e_i - e_j) \cdot \varphi = \varphi_i - \varphi_j$	quadratic form of a rank-one edge matrix the edge vector selects the potential difference
T3_sfi_edge_sensitivity	$\varphi^\top E_{ij} \varphi = (\varphi_i - \varphi_j)^2$	the SFI edge term
T4_sfi_term_nonneg	$0 \leq w \Rightarrow 0 \leq w(\varphi_i - \varphi_j)^2$	each SFI contribution is non-negative
T5_aggregate_nonneg	$\sum (\text{nonneg terms}) \geq 0$	SFI is a Dirichlet energy; the Laplacian form is PSD

Combined with the classical Hellmann–Feynman theorem $\partial \lambda_2 / \partial w_{ij} = \varphi_2^\top (\partial L / \partial w_{ij}) \varphi_2$ (Ghosh–Boyd 2006 [2], an analysis result we cite, not re-prove), T3 is exactly the closed form $\partial \lambda_2 / \partial w_{ij} = (\varphi_{2,i} - \varphi_{2,j})^2$. Lean/Mathlib was attempted; its toolchain binaries are distributed only as GitHub release assets, which the compute environment’s egress policy blocks with a hard 403, so Coq provides the equivalent kernel-checked guarantee. Reproduce with `coqc formal/sfi_edge_identity_Q.v`.

B Reproducibility

All stochastic steps are seeded; the exact configuration and commit hash accompany every reported number. Numeric verification: `python scripts/verify_sfi_identity.py` and `python`

`scripts/verify_math_rigor.py` (sympy symbolic, mpmath 40-digit, exact-rational, interval-certified, and cross-solver checks). Convergence: `python scripts/convergence_analyze.py roney`. Figures are regenerated by `scripts/make_paper_figures.py`. Datasets: Roney LA cohort (Zenodo 5801337) and UW/Boyle meshes (Dryad 10.5061/dryad.kkwh70sg0); both are public and de-identified.

References

- [1] M. Fiedler. Algebraic connectivity of graphs. *Czechoslovak Mathematical Journal*, 23(2):298–305, 1973.
- [2] A. Ghosh and S. Boyd. Growing well-connected graphs. In *Proc. 45th IEEE Conference on Decision and Control (CDC)*, 6605–6611, 2006.
- [3] L. M. Pecora and T. L. Carroll. Master stability functions for synchronized coupled systems. *Physical Review Letters*, 80(10):2109–2112, 1998.
- [4] M. Barahona and L. M. Pecora. Synchronization in small-world systems. *Physical Review Letters*, 89(5):054101, 2002.
- [5] W. Bomela, S. Wang, C.-A. Chou, and J.-S. Li. Real-time inference and detection of disruptive EEG networks for epileptic seizures. *Scientific Reports*, 10:8653, 2020.
- [6] H. Weyl. Das asymptotische Verteilungsgesetz der Eigenwerte linearer partieller Differentialgleichungen. *Mathematische Annalen*, 71(4):441–479, 1912.
- [7] C. Davis and W. M. Kahan. The rotation of eigenvectors by a perturbation. III. *SIAM Journal on Numerical Analysis*, 7(1):1–46, 1970.
- [8] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*, 44(3):837–845, 1988.
- [9] X. Sun and W. Xu. Fast implementation of DeLong’s algorithm for comparing the areas under correlated ROC curves. *IEEE Signal Processing Letters*, 21(11):1389–1393, 2014.
- [10] C. C. Mitchell and D. G. Schaeffer. A two-current model for the dynamics of cardiac membrane. *Bulletin of Mathematical Biology*, 65(5):767–793, 2003.
- [11] C. H. Roney et al. Predicting atrial fibrillation recurrence by combining population data and virtual cohorts of patient-specific left-atrial models. *Circulation: Arrhythmia and Electrophysiology*, 15(2):e010253, 2022. doi:10.1161/CIRCEP.121.010253. Data: Zenodo 5801337.
- [12] S. Bifulco, P. M. Boyle, et al. Predicting arrhythmia recurrence post-ablation in atrial fibrillation using explainable machine learning. *Communications Medicine*, 2025. doi:10.1038/s43856-025-01058-4. Data: Dryad 10.5061/dryad.kkwh70sg0.
- [13] I. M. Khurram et al. Magnetic resonance image intensity ratio, a normalized measure to enable interpatient comparability of left atrial fibrosis. *Heart Rhythm*, 11(1):85–92, 2014.
- [14] S. Kumar et al. Atrial fibrillation inducibility in the absence of structural heart disease or clinical atrial fibrillation: critical dependence on induction protocol, inducibility definition, and number of inductions. *Circulation: Arrhythmia and Electrophysiology*, 5(3):531–536, 2012. PMID 22528047.

- [15] J. Marquardt et al. Prognostic value of atrial fibrillation inducibility in patients without history of clinical atrial fibrillation. *Journal of Atrial Fibrillation*, 11(1):1837, 2018. PMID 30455834.
- [16] H. Oral et al. Inducibility of paroxysmal atrial fibrillation by isoproterenol and its relation to the mode of onset of atrial fibrillation. *Journal of Cardiovascular Electrophysiology*, 19(5):466–470, 2008. PMID 18266669.
- [17] S. Kawai et al. Predictive value of the induction test with atrial burst pacing with regard to long-term recurrence after ablation in persistent atrial fibrillation. *Journal of Arrhythmia*, 35(2):223–229, 2019. PMID 31007786.
- [18] X. Liu et al. Is atrial fibrillation noninducibility by burst pacing after catheter ablation associated with reduced clinical recurrence? A systematic review and meta-analysis. *Journal of the American Heart Association*, 9(14):e015260, 2020. PMID 32654581.